

assignment _adding_vectors ↕

⚠ This is a preview of the published version of the quiz

Started: Apr 29 at 7:30pm

Quiz Instructions

There are tutorials in my dropbox that cover vectors and adding vectors (in PDFs_SLIDES_videos_tutorials_problems). I highly recommend you to watch those videos if you are still struggling with vectors.

Also. Review textbook. vectors are very important for engineering.

You can also use the simulation:

https://phet.colorado.edu/sims/html/vector-addition/latest/vector-addition_en.html 
(<http://simulation%20vectors.%20Use%20it>)

No tutorial in the HW folder for this assignment

Use a cartesian coordinate system. tails of vectors are tied up to the origin.



Question 1 10 pts

The x component of vector is 5.3 units, and its y component is -2.3 units. The angle that vector makes with the $+x$ -axis is closest to



340°.



250°.



23°.



110°.



160°.



Question 2 10 pts

The components of vectors $\vec{B}$ and $\vec{C}$ are given as follows:

$$B_x = -9.2 \quad C_x = -4.5$$

$$B_y = -6.1 \quad C_y = 4.3$$

The angle (less than 180°) between vectors $\vec{B}$ and $\vec{C}$ is closest to

hint: You can trace the vector in a X-Y coordinate system. Attach the tails to the origin. $\vec{B}$ should be in the third quadrant and $\vec{C}$ in the 4th quadrant. Use your trig skill.

If you know about the dot product, you can use it instead.

☐

170° .

☐

103° .

☐

84° .

☐

10° .

☐

77° .

☐

Question 3 10 pts

The components of vectors $\vec{A}$ and $\vec{B}$ are given as follows:

$$A_x = 7.6 \quad B_x = -5.1$$

$$A_y = -9.2 \quad B_y = -6.8$$

What is the magnitude of the vector difference $\vec{C} = \vec{B} - \vec{A}$?

hint: $C_x = B_x - A_x$ and $C_y = B_y - A_y$. Plot (x_C, y_C) in the coordinate system. It's the head of the vector $\vec{C}$. Its tail is the origin.

☐

16

☐

13

☐

3.5

☐

3.4

☐

170



Question 4 10 pts

If vector A has components $A_x = -3.0$ lb and $A_y = -4.0$ lb, and vector B has components $B_x = 3.0$ lb and $B_y = -8.0$ lb, what is the magnitude of vector $C = A - B$?

☐

16 lb

☐

7.2 lb

☐

140 lb

☐

13 lb



Question 5 10 pts

Vector M = 4.00 m points eastward and vector N = 3.00 m points southward. The resultant vector M + N is given by

☐5.00 m at an angle of 26.6° south of east.☐5.00 m at an angle of 53.1° south of east.☐5.00 m at an angle of 18.4° south of east.☐5.00 m at an angle of 36.9° south of east.☐5.00 m at an angle of 71.6° south of east.

Question 6 10 pts

Vector A has a magnitude of 6.0 m and points 30° north of east. Vector B has a magnitude of 4.0 m and points 30° east of north. The resultant vector A + B is given by

☐9.7 m at an angle of 42° north of east.☐0.70 m at an angle of 42° north of east.☐1.1 m at an angle of 42° north of east.☐2.0 m at an angle of 42° north of east.☐14 m at an angle of 42° north of east.

Question 7 10 pts

Vector A has a magnitude of 4.0 m and points 30° south of east. Vector B has a magnitude of 2.0 m and points 30° north of west. The resultant vector $A + B$ is given by

☐

10.0 m at an angle 30° south of east.

☐

2.0 m at an angle 60° south of east.

☐

1.0 m at an angle 30° east of south.

☐

10.0 m at an angle 60° east of south.

☐

2.0 m at an angle 30° south of east.



Question 8 10 pts

Three vectors, S, T, and U, have the components shown in the table. What is the magnitude of the resultant (sum R) of these three vectors?

hint: Add the x-components to get the R_x and add the y-components to get the R_y .

Trace R then find the magnitude.

	x component	y component
S	3.50 m	-4.50 m
T	2.00 m	0.00 m
U	-5.50 m	2.50 m

☐

11.1 m

☐

5.50 m

☐

2.00 m

☐

13.0 m

☐

7.00 m



Question 9 10 pts

A plane has an airspeed of 200 m/s northward, and is in a wind of 50.0 m/s to the west. The plane's speed relative to the ground is

hint: add vectors. $V_{\text{plane/wind}} + V_{\text{winf/ground}}$. Trace the vectors. Find the magnitdue.



250 m/s.



150 m/s.



206 m/s.



200 m/s.



Question 10 10 pts

A plane moving 200 m/s horizontally fires a projectile with speed 50 m/s in a forward direction 30.0° below the horizontal. At that instant, what is the speed of the projectile with respect to a stationary observer on the ground?

hint: Add the vectors. One vector is horizontal and one has a direction -30



293 m/s



268 m/s



245 m/s



250 m/s



Question 11 5 pts

bonus

Object A has a position as a function of time given by $\vec{r}_A(t) = (3.00 \text{ m/s})t\hat{i} + (1.00 \text{ m/s}^2)t^2\hat{j}$. Object B has a position as a function of time given by $\vec{r}_B(t) = (4.00 \text{ m/s})t\hat{i} + (-1.00 \text{ m/s}^2)t^2\hat{j}$. All quantities are SI units. What is the distance between object A and object B at time $t = 3.00 \text{ s}$?

hint: 1) First find the coordinates of A (x_A, y_A) and B(x_B, y_B) at time 3s ->

for example $x_A = 9\text{m}$

2) Then find the displacement AB [($x_B - x_A$) , ($y_B - y_A$)].

3) Plot the point [$x_B - x_A$, $y_B - y_A$] in a coordinate system.

Connect origin to the point. This is the vector AB.

4) use pythagorean theorem ..for magnitude

☐

15m

☐

3.46m

☐

29.8m

☐

34.6m

☐

18.3m

Not saved

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